NETFLIX RELEASE-YEAR FORECASTING

Auspify Internship - Task 4 Report

1. Objective

The objective of this task was to analyze Netflix content release trends by year and build a forecasting model to estimate future content levels.

2. Task Workflow

Step 1: Prepare release-year data.
Step 2: Analyze yearly content trends.
Step 3: Build forecasting models.
Step 4: Visualize future predictions.
Step 5: Interpret results.

3. Data Preparation

The Netflix dataset was cleaned using the release_year field. Release years were converted to numeric values, records with missing years were removed, and the number of titles released in each year was calculated. This produced a yearly time series suitable for trend analysis and forecasting.

4. Trend Analysis

The yearly content counts were grouped by release year and visualized using a line chart. The analysis was used to identify changes in the volume of Netflix content over time and to determine the year with the highest number of titles in the dataset.

5. Forecasting Model

A Linear Regression model was trained using release year as the input feature and yearly content count as the target variable. The dataset was divided into training and testing sets, and the model was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

6. Future Predictions

The trained model was used to forecast Netflix content counts for the five years immediately following the latest release year available in the dataset. The predictions were plotted together with the historical trend to make the expected direction of change easier to interpret.

7. Evaluation and Interpretation

MAE and RMSE were calculated to measure prediction error; lower values indicate better predictive accuracy. The forecast direction was determined by comparing the first and last predicted values. These results provide a baseline view of how release activity may continue based on the historical pattern captured by the model.

8. Conclusion

Task 4 successfully applied time-series preparation, trend analysis, regression forecasting, model evaluation, and data visualization to Netflix release-year data. The project provided practical experience in forecasting and predictive analytics. The Linear Regression model serves as a simple baseline and could be improved by testing models designed specifically for time-series forecasting and by incorporating additional explanatory variables.

9. Skills Demonstrated

Python, Pandas, Scikit-learn, time series analysis, forecasting, Linear Regression, MAE, RMSE, predictive analytics, and data visualization.

Project Outcome: A complete baseline forecasting workflow for Netflix content release trends, including cleaned yearly data, trend analysis, a forecasting model, evaluation metrics, and future prediction visualization.